

Web-based Supplementary Materials for One point per cluster spatially balanced sampling

B. L. Robertson^{1,*} and C. J. Price¹

¹School of Mathematics and Statistics, University of Canterbury, Christchurch, New Zealand

*email: blair.robertson@canterbury.ac.nz

Response Functions

The following functions were used to define the response variables in Section 4.3.

- **Corrugated Plane** (Grafström, Lundström and Schelin 2012; Robertson et al. 2013, 2018, 2022):

$$f(\mathbf{x}) = 3(x_1 + x_2) + \sin(6(x_1 + x_2)),$$

with $\tau \approx 3001$ for the regular population and $\tau \approx 2891$ for the aggregated population.

- **Peak function** (Robertson et al. 2018, 2022):

$$\begin{aligned} f(\mathbf{x}) = & 3(4 - 6x_1)^2 \exp(-(6x_1 - 3)^2 - (6x_2 - 2)^2) \\ & - 10(0.2(6x_1 - 3) - (6x_1 - 3)^3 - (6x_2 - 3)^5) \exp(-(6x_1 - 3)^2 - (6x_2 - 3)^2) \\ & - \frac{1}{3} \exp(-(6x_1 - 2)^2 - (6x_2 - 3)^2), \end{aligned}$$

with $\tau \approx 7378$ for the regular population and $\tau \approx 7264$ for the aggregated population.

- **Bird function** (Robertson et al. 2018, 2022):

$$\begin{aligned} f(\mathbf{x}) = & \frac{1}{20} \left[(12x_1 - 12x_2)^2 + \exp[(1 - \sin(12x_1 - 6))^2] \cos(12x_2 - 6) \right. \\ & \left. + \exp[(1 - \cos(12x_2 - 6))^2] \sin(12x_1 - 6) \right], \end{aligned}$$

with $\tau \approx 7185$ for the regular population and $\tau \approx 6671$ for the aggregated population.

References

- [1] Grafström, A., Lundström, N. L. P. and Schelin, L. (2012). Spatially balanced sampling through the pivotal method. *Biometrics* **68**, 514–520.
- [2] Robertson, B. L., Brown, J. A., McDonald, T. and Jaksons, P. (2013). BAS: Balanced acceptance sampling of natural resources. *Biometrics* **3**, 776–784.
- [3] Robertson, B. L., McDonald, T., Price, C.J. and Brown, J. A. (2018). Halton iterative partitioning: spatially balanced sampling via partitioning. *Environmental and Ecological Statistics* **3**, 305–323.
- [4] Robertson, B. L., Ozturk, O., Kravchuk, O. and Brown, J. A. (2022). Spatially Balanced Sampling with Local Ranking. *Journal of Agricultural, Biological and Environmental Statistics* **27**, 622–639.